# Aayush Parikh

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## Education

**Arizona State University** — *Tempe, AZ*
*Master of Science in Computer Science, GPA: 3.93* — August 2024 – May 2026

**University of Mumbai** — *Mumbai, India*
*Bachelor of Science in Computer Science, Honors in Intelligent Computing, GPA: 3.9* — December 2020 – June 2024

## Certifications

**AWS Certified Cloud Practitioner** | *Amazon Web Services* — October 2025 – October 2028

## Experience

**Brinks Home** — *Seattle, WA (Remote)*
*Software Developer* — June 2026 – Present
*Software Engineering Intern* — May 2025 – May 2026

- Engineered a concurrency-safe event-handling pattern for a distributed order-activation pipeline (C#, RabbitMQ), resolving race conditions across asynchronous billing-activation calls with zero added overhead.
- Constructed fault-tolerant backend REST APIs utilizing RabbitMQ to decouple inventory-tracking and event notification workloads; sustained high traffic profiles with zero message loss via asynchronous queue processing.
- Developed a low-latency Vue-based checkout flow, decreasing time-to-interact by 25% and cart abandonment by 12%; utilized Claude Code to rapidly scaffold TypeScript TestCafe suites to ensure reliability and consistency.

**Acodesoft Technologies LLP** — *Mumbai, India*
*Software Engineering Intern* — June 2023 – September 2023

- Architected and deployed highly available Node.js microservices on AWS EC2 incorporating Auto Scaling Groups and optimized infrastructure compute capacity based on live traffic, maintaining a 99.9% platform availability baseline.
- Streamlined software delivery lifecycle by architecting CI/CD pipelines leveraging GitHub Actions, accelerating code-to-production speed by 40% and eliminating manual rollback errors.

**TechData Solutions** — *Mumbai, India*
*Software Developer Intern* — July 2022 – October 2022

- Designed a high-throughput C++ data ingestion and normalization layer for live streams, reducing end-to-end latency by 20% via memory access optimization and cache-friendly data structures.
- Augmented backend payload retrieval in multi-threaded environments by structuring custom state-caching mechanisms, resulting in a measurable reduction in P99 tail latency and system jitter.

## Projects

**Distributed Real-Time Task Coordinator** | *Java, Spring Boot, Project Loom, Redis*

- Created a distributed task coordination service in Java, employing Virtual Threads (Project Loom) to manage 20,000+ concurrent background tasks with 50% less memory consumption than traditional thread pools.
- Devised a centralized locking and state management system using Redis, ensuring tasks execute exactly once across a multi-node cluster even during unexpected server failovers.

**Advanced Shell** | *C++, POSIX, Docker, AWS SDK, CloudWatch*

- Programmed a POSIX-compliant shell in C++ handling process scheduling and inter-process communication (IPC) via pipes and signals to manage low-level kernel interactions and resource allocation.
- Embedded the AWS SDK for C++ into shell runtime to stream custom execution telemetry to CloudWatch, enabling real-time observability of background jobs and instantaneous alerting for resource anomalies.

**Serverless Real-Time Data Pipeline** | *AWS Kinesis, Lambda, DynamoDB, Terraform*

- Architected an event-driven workflow utilizing AWS Kinesis to ingest mock financial data streams, triggering AWS Lambda functions to normalize and persist records to DynamoDB with sub-second latency during load testing.
- Accelerated the provisioning of cloud infrastructure by leveraging Claude Code to generate base Terraform scripts, optimizing configuration workflows and cutting manual setup time by 80%.

## Technical Skills

**Languages & Tools**: Python, C, C++, C#, JavaScript, Java, SQL, TypeScript, Swift, Kotlin, Go, Ruby, .NET, Vercel
**Databases & Platforms**: MongoDB, MySQL, Docker, Kubernetes, Claude Code, Git, AWS, NoSQL, Redis
**Frameworks & Libraries**: React.js, React Native, Node.js, Express.js, Vue.js, Bootstrap, TensorFlow, Redux, Keras